

Hein Zarni Aung

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EDUCATION

Stony Brook University

Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Systems Fundamentals I & II, Algorithms, Programming Abstractions

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Dept. of Computer Science, Stony Brook University

Sep 2025 – Dec 2025

PI: Abisheka Pitumpe | Team: 2

- Built a real-time pipeline to monitor newly issued SSL/TLS certificates and flag fraudulent domains, using Selenium to automate scraping and Python to filter, triage, and validate flagged sites; collaborated with a faculty PI in a two-person research team to design the detection logic.

INTERNSHIPS

Software Engineering Intern — Clustr

Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows, like/dislike interactions, and activity feed endpoints.

Software Engineering Intern — Prenora Dynamics LLC

Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors to schedule and host lessons in one platform, and built a seed-data workflow to streamline debugging and local development.

PROJECTS

Diary Handwriting Analysis (Python, PyTorch Hugging Face Transformers, OpenCV)

- Fine-tuned Microsoft's TrOCR handwriting recognition model with a custom OpenCV line-segmentation preprocessing pipeline (grayscale conversion, binary thresholding, horizontal ink-projection profiling), reaching ~87% reliable page transcription on personal diary pages.
- Pivoted to DANIEL, a segmentation-free machine learning architecture that transcribes full pages directly and jointly extracts named entities, enabling structured topic and mood-trend analytics across diary entries over time; currently in progress.

TuneAcademy (React, NumPy, Essentia, Firestore)

- Built the AI/audio-analysis pipeline for an app matching music students with instructors, extracting pitch, rhythm, and tone features with Essentia/NumPy, then pivoting from a planned ML model to rule-based scoring under hackathon time and compute constraints, delivering reliable detection within the 36-hour build window.
- Set up a CI/CD pipeline using GitHub Actions to lint and test the codebase, then auto-build and deploy the app to Cloudflare Workers.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Dept. of Computer Science, Stony Brook University

Aug 2025 – Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations; created original review materials adopted for subsequent semesters and led midterm review sessions reaching 100+ students combined in-person and via Zoom.

SKILLS

Languages: Python, C, Java, JavaScript, TypeScript

Tools: Linux, Git, Docker, PM2, Selenium

Backend: Node.js, Express.js, FastAPI

Frontend: React, React Native, Next.js

Databases & Cloud: MongoDB, PostgreSQL, Firebase/Firestore, AWS, GCP